

ACTL3142 Tutorial

Week 2: Simple Linear Regression

Nick Guo

School of Risk and Actuarial Studies, UNSW Sydney

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Recap from Last Week

Key ideas from Week 1:

- ▶ Statistical learning: relate $\mathbf{X}$ to Y via $Y = f(\mathbf{X}) + \varepsilon$
- ▶ Supervised vs unsupervised; prediction vs inference
- ▶ Parametric vs non-parametric; flexibility vs interpretability

Bias-variance tradeoff:

- ▶ More flexible $\Rightarrow$ less bias, more variance
- ▶ Less flexible $\Rightarrow$ more bias, less variance
- ▶ Best model achieves an optimal balance

Today we study **simple linear regression**: a parametric, supervised method that sits at the *inflexible but interpretable* end of the spectrum.

Simple Linear Regression: The Model

Relate a *single* predictor X to a **quantitative** response Y :

$$\mathbf{y} = \beta_0 + \beta_1 \mathbf{x} + \varepsilon$$

Interpreting the coefficients:

- ▶ β_0 (intercept): predicted value of Y when $X = 0$
- ▶ β_1 (slope): average change in Y per unit increase in X

What does “linear” actually mean?

Linear regression is **linear in the parameters** β_0, β_1 , not necessarily in X . For example, $y = \beta_0 + \beta_1 x^2 + \varepsilon$ is still a linear regression model (this idea comes back later in the course).

Assumptions on the Errors

To estimate β_0 and β_1 , we need assumptions on ε .

Weak assumptions (for OLS):

- ▶ $\mathbb{E}[\varepsilon_i | \mathbf{x}] = 0$, $\text{Var}(\varepsilon_i | \mathbf{x}) = \sigma^2$, $\text{Cov}(\varepsilon_i, \varepsilon_j | \mathbf{x}) = 0$ for $i \neq j$
- ▶ In words: zero mean, constant variance, uncorrelated errors
- ▶ Enough to get point estimates and standard errors

Strong assumptions (for MLE):

- ▶ $\varepsilon_i | \mathbf{x} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$
- ▶ Adds normality and independence on top of the weak assumptions
- ▶ Needed for confidence intervals, hypothesis tests, and prediction intervals

*Tip: under strong assumptions, OLS and MLE give the same $\hat{\beta}$'s. The difference is what you can **do** with them (inference requires the stronger set).*

Least Squares Estimates (LSE)

The idea: find the $\hat{\beta}_0, \hat{\beta}_1$ that minimise the **residual sum of squares**:

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Setting partial derivatives to zero gives:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

*Intuition: OLS finds the line that minimises the sum of squared **vertical** distances from each data point to the line.*

Properties and Standard Errors

Under the weak assumptions, the LSE are **unbiased**:

$$\mathbb{E}[\hat{\beta}_0 \mid \mathbf{x}] = \beta_0 \quad \text{and} \quad \mathbb{E}[\hat{\beta}_1 \mid \mathbf{x}] = \beta_1$$

Their variances are:

$$\text{SE}(\hat{\beta}_1)^2 = \text{Var}(\hat{\beta}_1 \mid \mathbf{x}) = \frac{\sigma^2}{S_{xx}}$$

$$\text{SE}(\hat{\beta}_0)^2 = \text{Var}(\hat{\beta}_0 \mid \mathbf{x}) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

We estimate σ^2 with $s^2 = \frac{\text{RSS}}{n-2}$.

Why $n - 2$? Think of it as a budget: you start with n independent pieces of information. Estimating $\hat{\beta}_0$ and $\hat{\beta}_1$ “spends” 2, leaving $n - 2$ degrees of freedom. More on this next.

Partitioning Variability

Total variability in Y decomposes as:

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{\text{TSS}} = \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{\text{RSS}} + \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{\text{MSS}}$$

- ▶ **TSS** (total sum of squares): how much Y varies overall
- ▶ **RSS** (residual sum of squares): variation left *unexplained* by the model
- ▶ **MSS** (model sum of squares): variation *explained* by the model

$$R^2 = \frac{\text{MSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

R^2 : the proportion of variability in Y explained by X .

In SLR, $R^2 = r_{xy}^2$ (the square of the sample correlation).

Degrees of Freedom as a Budget

You collect n data points. Think of this as a **budget** of n independent pieces of information. Every parameter you estimate “spends” one.

Quantity	D.F.	Why?
Total	$n - 1$	Computing TSS requires $\bar{y}$, costing 1
Error	$n - 2$	Computing RSS requires $\hat{\beta}_0$ and $\hat{\beta}_1$, costing 2
Model	1	The slope β_1 is the single parameter explaining variation

Notice: $(n - 1) = (n - 2) + 1$. The degrees of freedom partition just like the sums of squares do.

This generalises: with p predictors, Error D.F. = $n - p - 1$, Model D.F. = p .

ANOVA Table

Source	D.F.	SS	MS	F
Regression	1	MSS	$\frac{MSS}{1}$	$\frac{MSS/1}{RSS/(n-2)}$
Error	$n - 2$	RSS	$\frac{RSS}{n - 2}$	
Total	$n - 1$	TSS		

Tip: the Mean Square for Error is just s^2 , and the F-ratio tests whether the regression explains significantly more variation than random noise. In SLR, $F = t_{\beta_1}^2$.

Pitfalls with R^2

Common misconceptions:

- ▶ High R^2 does **not** mean the model is “good” or correct
 - ▶ A non-linear relationship can have a high R^2 with a linear fit but the model may be badly misspecified
- ▶ Low R^2 does **not** mean X is unrelated to Y
 - ▶ If σ^2 is large, R^2 will be low even if $\beta_1 \neq 0$
- ▶ R^2 can only increase (never decrease) when you add more predictors
 - ▶ This is why adjusted R^2 exists (we'll see this in MLR)

Always check residual plots alongside R^2 . A number alone doesn't tell the full story.

Q4: ANOVA Table

4. [Solution](#) Complete the following ANOVA table for a simple linear regression with $n = 60$ observations:

Source	D.F.	Sum of Squares	Mean Squares	F-Ratio
Regression	_____	_____	_____	_____
Error	_____	_____	8.2	
Total	_____	639.5		

★ [Solution](#) Consider a fitted simple linear regression model (via least squares) with estimated parameters $\hat{\beta}_0$ and $\hat{\beta}_1$. Let x_0 be a new (but known) observation (i.e., not in the training set used to estimate the parameters). Your prediction of the response for x_0 would be $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Prediction and Its Variance

For a new observation x_0 , our prediction of the mean response is:

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

Variance of the predicted **mean**:

$$\text{Var}(\hat{y}_0 \mid \mathbf{x}) = \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}} \right)$$

Variance of a predicted **individual** response $\tilde{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \varepsilon_0$:

$$\text{Var}(\tilde{Y}_0 \mid \mathbf{x}) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}} \right)$$

Notice: both variances are smallest when $x_0 = \bar{x}$ and grow as x_0 moves away from the centre of the data. This is why prediction bands have a “bowtie” shape.

Q5: Variance of Prediction

5. a. Show that the variance of this prediction is

$$\mathbb{V}(\hat{y}_0|\mathbf{x}) = \left(\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}} \right) \sigma^2.$$

Note you may use as given the expressions for the (co)-variances of $\hat{\beta}_0, \hat{\beta}_1$ given in the lecture.

- b. Notice how $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$ does not contain an “error term” (because the only reasonable prediction of the error term is 0). But this means that the variance of $\hat{y}_0$ computed above should be viewed as the variance of the “predicted mean” of the new observation Y_0 . What is the variance of the **predicted individual response** $\hat{Y}_0$? **Hint:** $\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \epsilon_0$.

Confidence Intervals

Under the **strong assumptions**, we can build CIs and do hypothesis tests.

CI for a coefficient (e.g. β_1):

$$\hat{\beta}_1 \pm t_{1-\alpha/2, n-2} \cdot \widehat{SE}(\hat{\beta}_1)$$

CI for the **mean response** at x_0 :

$$\hat{y}_0 \pm t_{1-\alpha/2, n-2} \cdot s \sqrt{\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}}}$$

Prediction interval for an **individual** response at x_0 :

$$\hat{y}_0 \pm t_{1-\alpha/2, n-2} \cdot s \sqrt{1 + \frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}}}$$

The prediction interval is always wider. Make sure you know which one a question is asking for!

Hypothesis Tests

To test $H_0: \beta_1 = \tilde{\beta}_1$, compute:

$$t = \frac{\hat{\beta}_1 - \tilde{\beta}_1}{\widehat{\text{SE}}(\hat{\beta}_1)} \sim t_{n-2} \quad \text{under } H_0$$

Alternative H_1	Reject H_0 if
$\beta_1 \neq \tilde{\beta}_1$ (two-sided)	$ t > t_{1-\alpha/2, n-2}$
$\beta_1 > \tilde{\beta}_1$ (one-sided)	$t > t_{1-\alpha, n-2}$
$\beta_1 < \tilde{\beta}_1$ (one-sided)	$t < -t_{1-\alpha, n-2}$

Most commonly: $H_0: \beta_1 = 0$ (is there any linear relationship?).

Tip: the p-value in R's `summary(lm(...))` output is for the two-sided test $H_0: \beta_j = 0$. For a one-sided test, halve it if $\hat{\beta}_j$ falls on the side H_1 claims; otherwise the one-sided p-value is $1 - p/2$.

Reading R Output

When you run `summary(lm(y ~ x))` in R, you get:

- ▶ **Coefficients table:** $\hat{\beta}$'s, their standard errors, t -stats, and p -values
- ▶ **Residual standard error:** this is $s = \sqrt{s^2}$ (on $n - 2$ d.f.)
- ▶ **Multiple R-squared:** R^2
- ▶ **F-statistic:** the F -ratio from the ANOVA table

You also get an **Analysis of Variance** table from `anova(lm(y ~ x))`.

Tip: for Q6, you'll need to read a printout like this. The "Coef" column gives $\hat{\beta}$'s, and "SE Coef" gives $\widehat{SE}(\hat{\beta})$. Everything else follows.

Q6: EPS vs STKPRICE

★ [Solution](#) Suppose you are interested in relating the accounting variable EPS (earnings per share) to the market variable STKPRICE (stock price). Then, a regression equation was fitted using STKPRICE as the response variable with EPS as the predictor variable.

Following is the computer output from your fitted regression. You are also given that: $\bar{x} = 2.338$, $\bar{y} = 40.21$, $s_x = 2.004$, and $s_y = 21.56$. (Note that: $s_x^2 = \frac{SS_x}{n-1}$ and $s_y^2 = \frac{SS_y}{n-1}$)

Regression Analysis

The regression equation is

STKPRICE = 25.044 + 7.445 EPS

Predictor	Coef	SE Coef	T	p
Constant	25.044	3.326	7.53	0.000
EPS	7.445	1.144	6.51	0.000

Analysis of Variance

SOURCE	DF	SS	MS	F	p
Regression	1	10475	10475	42.35	0.000
Error	46	11377	247		
Total	47	21851			

- Compute s and R^2 .
- Calculate the correlation coefficient of EPS and STKPRICE.
- Estimate the STKPRICE given an EPS of \$2. Provide 95% confidence intervals on both the "predicted mean" and the "predicted individual response" for STKPRICE (these are defined in an [previous question](#)). You may assume that the predicted mean and predicted response are approximately Normal (this is not *exactly* true, even under the strong assumptions, but a reasonable assumption if the sample size n is decently large). Comment on the difference between those two confidence intervals.
- Provide a 95% confidence interval for the slope coefficient β .
- Describe how you would check if the errors have constant variance.
- Perform a test of the significance of EPS in predicting STKPRICE at a level of significance of 5%.
- Test the hypothesis $H_0 : \beta = 6$ against $H_a : \beta > 6$ at a level of significance of 5%.

Q3: University Exam Data

3. ★ [Solution](#) A university wishes to analyse the performance of its students on a particular degree course. It records the scores obtained by a sample of 12 students at the entry to the course, and the scores obtained in their final examinations by the same students. The results are as follows:

Student	A	B	C	D	E	F	G	H	I	J	K	L
Entrance exam score x (%)	86	53	71	60	62	79	66	84	90	55	58	72
Final paper score y (%)	75	60	74	68	70	75	78	90	85	60	62	70

$$\sum x = 836, \sum y = 867, \sum x^2 = 60,016, \sum y^2 = 63,603, \sum (x - \bar{x})(y - \bar{y}) = 1,122.$$

- Calculate the fitted linear regression equation of y on x .
- Under the strong assumptions, it can be shown that

$$\frac{s^2}{\sigma^2/(n-2)} \sim \chi_{n-2}^2,$$

where s^2 is the unbiased estimate of σ^2 in simple linear regression. Knowing this, calculate an estimate of the variance σ^2 and obtain a 90% confidence interval for σ^2 , for the above data.

- By considering the slope parameter, formally test whether the data is positively correlated.
- Calculate the proportion of variance explained by the model. Hence, comment on the fit of the model.

Thanks for coming! See you next week.

